

## **MPA vs SPA (Short Answer):**

- **MPA (Multi-Page Application):**

Each request loads a new page from the server (full page reload). Rendering is server-side. Best for simple apps and SEO-focused websites.

- **SPA (Single-Page Application):**

Only one page is loaded, and content updates dynamically without reloading. Rendering is client-side using JavaScript. Best for fast and interactive applications.

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## **Best approach for each project:**

1. **Management System:**

**MPA** — simple CRUD operations, no complex interactivity needed.

2. **Dashboard / Admin Panel:**

**SPA** — requires fast interaction, real-time updates, and smooth UX.

3. **E-commerce Website:**

**Hybrid (MPA + SPA)** —

MPA for SEO-friendly pages (products, landing pages) + SPA for interactive features (cart, filtering).

## **Difference between REST and gRPC:**

### **REST (Representational State Transfer):**

- Uses standard HTTP methods like GET, POST, PUT, DELETE.
  - Data is usually sent in **JSON format** (human-readable).
  - Easy to use, widely supported, and simple to debug.
  - Slower compared to gRPC because of text-based data and HTTP overhead.
  - Best for public APIs and web/mobile applications.
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### **gRPC (Google Remote Procedure Call):**

- Uses **HTTP/2** and sends data in **binary format (Protocol Buffers)**.
- Much faster and more efficient than REST.
- Supports real-time communication, streaming, and bi-directional calls.
- Strongly typed (requires .proto files).
- More complex to implement and debug.